

Unit 12, Lesson 13: Writing and Solving One-Step Equations in One Variable

Benchmarks

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Learning Targets

- ☐ I can write and solve one-step equations in one variable within a real-world context using multiplication and division.
- ☐ I can solve one-step equations in one variable using addition and subtraction.

12.13.1 Warm-Up: One-Step Equations

1. Fill in each blue box to make the equations true. Use only digits 1 to 9, at most one time each.

$$\boxed{} + a = \boxed{}$$

$$\boxed{} b = \boxed{}$$

$$c - \boxed{} = \boxed{}$$

$$a = \boxed{}, b = \boxed{}, c = \boxed{}$$

12.13.2 Collaboration: Relating Real-World Contexts to Equations

1. Four equations are shown.

$$4x = 32$$
$$\frac{x}{32} = 4$$

$$32x = 4$$
$$\frac{x}{4} = 32$$

Discuss with your partner which of the equations correctly represents each situation below, where x represents the unknown quantity. Then, write the matching equation in the space provided.

Situation 1	Situation 2
At a restaurant, 4 friends decided to split the bill evenly. If each person paid \$32, what was the total bill?	A manatee has a total of 32 teeth. If a manatee has 4 rows of teeth with the same number of teeth in each row, how many teeth are in each row?
Equation:	Equation:

12.13.3 Guided Instruction: Solving One-Step Equations Representing Real-World Contexts Using Multiplication and Division

1. A town's total budget for firefighters' wages is \$320,000. Wages are calculated at \$40,000 per firefighter.
 - a. Draw a bar model that can be used to determine how many firefighters the town can afford to employ.
 - b. Write an equation that can be used to find w , the number of firefighters the town can employ.
 - c. What is the value of w ?
 - d. Explain what the solution means in the context of the problem.

2. There are 4 flooring companies submitting bids on a new housing development. Each company has 11 different flooring samples.

Dino said, "To find the total number of samples, s , being submitted, set up the equation $s = 4 \cdot 11$."

Dino's partner Callahan shared that he used the equation $\frac{s}{4} = 11$ to find the total number of samples submitted.

Bristol shouted across the classroom that she wrote the equation, $\frac{s}{11} = 4$.

- a. Discuss with your partner which student — Dino, Callahan, or Bristol — wrote the correct equation to solve for the total number of samples submitted.
- b. Solve one of the equations and then use a different equation to check your work.
- c. Explain how the answer makes sense in terms of the context.

12.13.4 Guided Practice: Solving Multiplication and Division Equations for Real-World Contexts

For each situation, write an equation that can be used to determine the unknown value. Then, solve the equation. Be sure to check your solution.

1. It takes 5 yards of fabric to make a Native American ceremonial dress.
- a. Write a division equation and a multiplication equation that can be used to determine how many yards, d , of fabric are needed to make 17 ceremonial dresses.

Division Equation	Multiplication Equation

- b. Solve either of the equations for d . Use the other equation to check your work.
2. Drake is selling candy bars for a school fundraiser. Each candy bar sells for \$3.00.
- a. Write a division equation and a multiplication equation that can be used to determine how many candy bars, c , he sold if he raised \$399.

Division Equation	Multiplication Equation

- b. Solve either of the equations for c . Use the other equation to check your work.
3. Discuss with your partner how creating the two different equations in Questions 1 and 2 demonstrate inverse relationships.

12.13.5 Try It: Mixed Review

Select all of the equations that represent each situation.

1. One Valencia orange tree can produce 275 oranges. A crate contains 25 oranges.
 - a. How many crates can one Valencia orange tree fill up? Select all that apply.
 - ☐ $\frac{x}{25} = 275$
 - ☐ $275 = 25x$
 - ☐ $275x = 25$
 - ☐ $\frac{x}{275} = 25$
 - ☐ $\frac{275}{x} = 25$
 - ☐ $\frac{275}{25} = x$
 - b. Pick one of the equations you selected and solve the problem. Show your reasoning.

2. Aliyah and Tiana want to bake 3 carrot cakes. Each carrot cake needs 2 cups of salted butter.
 - a. How much butter will they need to bake all 3 cakes? Select all that apply.
 - ☐ $\frac{x}{2} = 3$
 - ☐ $2x = 3$
 - ☐ $2(3) = x$
 - ☐ $3x = 2$
 - ☐ $6x = 2$
 - ☐ $\frac{x}{3} = 2$
 - ☐ $\frac{x}{3} = 6$

- b. Select one of the equations you selected and solve the problem. Show your reasoning.

3. Solve each equation for the variable. Verify your answers.

a. $5x = 230$	b. $y + 4 = -2$
c. $-16 + d = -26$	d. $r - 96 = 103$
e. $\frac{k}{4} = -51$	f. $-156 = -13w$

12.13.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning objective.



I need help.



I got it.



I could help
someone else.

Determine the value of an unknown decimal or fraction in an equation.	  
Translate a written description into an algebraic expression.	  
Write a description within a real-world context given an algebraic expression.	  
Write and solve one-step equations in one variable using addition and subtraction.	  
Write and solve one-step equations in one variable using multiplication and division.	  

